

Hyeju Shin

☎ (+82)10-5785-3367 | ✉ hyeju.shin@etri.re.kr | 🌐 hyeju99.github.io

Research Experience

- Electronics and Telecommunications Research Institute (ETRI)** Sep. 2024 – Present
Researcher at [Edge Computing Application Service Lab.](#) (Director: Dr. Jaemin Kim)
Gwangju, South Korea
- Researched Object Tracking models with Re-ID for CCTV environments like daycare centers and stores.
 - Provided vision services using Raspberry Pi and Hailo-8, with plans to use NPU chips from small enterprises.
- AISeed** Mar. 2024 – Sep. 2024
Researcher at Research Center
Gwangju, South Korea
- Established a real-time OCR solution for medical AI applications using PyTorch.
 - Standardized company-wide coding conventions and introduced GitHub templates for enhanced project management.
- Electronics and Telecommunications Research Institute (ETRI)** Nov. 2023 – Feb. 2024
Research Intern at [Edge Computing Application Service Lab.](#) (Director: Dr. Sung Chang Kim)
Gwangju, South Korea
- Developed and optimized a heterogeneous Federated Learning framework using Jetson Orin Nano/Xavier and K8S.
 - Accelerated pose-estimation inference on Jetson Nano with Hailo-8 and ONNX framework.
- Institut Polytechnique de Paris (IP Paris)** Jul. 2023 – Jul. 2023
Short-term Visiting Researcher at [RMS research group](#) (Advisor: Prof. Keun-Woo Lim)
Palaiseau, France
- Presented Digital Twin Network (DTN) technology [PPT] and facilitated discussions on on-device learning.
 - Assisted in formalizing a Memorandum of Understanding (MoU) between CNU and Télécom Paris.

Education

- Chonnam National University** Mar. 2022 – Feb. 2024
M.S. in Electronics and Computer Engineering (Advisor: Prof. Jinsul Kim)
Gwangju, South Korea
- **Thesis:** Improving the Performance of MPNN-based Network Evaluation Model for Digital Twin Network
- Chonnam National University** Mar. 2017 – Feb. 2022
B.E. in Industrial Engineering & Computer and Information Engineering
Gwangju, South Korea
- **Thesis:** Development of a deep learning model for detecting illegal driving in shared mobility using YOLO (Advisor: Prof. Ji-Seung Nam) (🏆 [The Excellence Prize of Capstone Design Contest](#))

Selected Publications

- [1] **H. Shin**, C. Kim, R. Kim, H. Yoo, and J. Kim, "Rethinking small vlm quantization: From component-wise analysis to hardware-aware edge deployment," *2026 International Conference on Machine Learning Workshops (ICML Workshops)*, Jul. 2026, [Paper].
- [2] **H. Shin**, K. Jung, S. Yun, and J. Yun, "Fast fourier transform-based spectral and temporal gradient filtering for differential privacy," *2025 The Conference on Information and Knowledge Management Workshops (CIKM Workshops)*, Nov. 2025, [Paper].
- [3] J. Shim, **H. Shin**, J. Kim, and H. Ryu, "Digital twin network technology trends for autonomous networks," *Electronics and Telecommunications Trends*, vol. 40, no. 1, pp. 12–21, 2025, [Paper].
- [4] **H. Shin**, I. Aliyu, A. Isah, and J. Kim, "Improving the real-data driven network evaluation model for digital twin networks," in *2024 IEEE International Conference on Communications Workshops (ICC Workshops)*, [Paper], [PPT], Denver, CO, USA, Jun. 2024.
- [5] **H. Shin**, S. Oh, A. Isah, I. Aliyu, J. Park, and J. Kim, "Network traffic prediction model in a data-driven digital twin network architecture," *Electronics*, vol. 12, no. 18, Sep. 2023, [Paper].
- [6] **H. Shin**, S. Oh, S. Oh, K. Kim, and J. Kim, "A study on the federated learning system architecture in hybrid cloud environment," *Korea Knowledge Information Technology Society (KKITS)*, vol. 18, no. 2, Apr. 2023, [Paper].

- [7] M. S. Miah and **H. Shin***, "E-scooter detection in media using deep learning," *The Journal of Contents Computing*, vol. 4, no. 1, pp. 393–399, Jun. 2022, [\[Paper\]](#).
- [8] I. Aliyu et al., "Toward a dynamic tasks offloading and resource allocation for the metaverse in in-network computing," in *2023 Fourteenth International Conference on Ubiquitous and Future Networks (ICUFN)*, [\[Paper\]](#), Paris, France, Jul. 2023.

Selected Patents

- [1] J. Kim and **H. Shin**, *System, method and program for transacting power forecasting ai model based on blockchain*, KR Patent 10-2570617, [\[Google Patents\]](#), Aug. 2023.
- [2] J. Kim, D. Lee, A. Ashiquzzaman, Y. Kim, and **H. Shin**, *Method for predicting potential confirmator of epidemic through analysis of cctv video*, KR Patent 10-2265617, [\[Google Patents\]](#), Jun. 2021.

Teaching Experience

IoT Computing, Chonnam National University

Fall 2022, Spring 2023

- Delivered lectures and hands-on sessions on IoT computing concepts using Arduino Uno, focusing on sensor integration and device control.
- Guided students in implementing AI-based object and facial recognition projects on Raspberry Pi, utilizing pre-trained models for real-world applications.
- Created step-by-step troubleshooting guides for Linux CLI commands, file management, and resolving library version conflicts, ensuring a smooth learning experience.

Undergraduate Capstone Project

Detecting Illegal Driving of Shared E-Scooter | Python, YOLOv3, labellmg, Roboflow

[\[Wiki\]](#) [\[Dataset\]](#)

- Led the development of a system to detect illegal driving behaviors on shared e-scooters, such as riding without a helmet or with two people on one scooter.
- Collected and labeled images of shared e-scooters using *labellmg*, and augmented the dataset using *Roboflow*.
- Implemented a real-time e-scooter company classifier using YOLOv3 with pre-trained weights from the COCO dataset.

Scholarships & Awards

BK21 Scholarship at BK21 Four IoT Research Center \$15,000	Mar. 2022 – Dec. 2023
TA Scholarship: IoT Computing at CNU \$2,400	Fall 2022, Spring 2023
Research Incentive for Graduate Student at BK21 Four IoT Research Center \$144	Aug. 2022, Feb. 2023
The Excellence Paper Prize at Korea Digital Contents Society (Domestic Conference)	Nov. 2022
TA Scholarship: Linux system and practices at CNU \$1,200	Spring 2022
Tuition Scholarships at CNU \$13,693	Mar. 2017 – Feb. 2022
TA Scholarship: Data Structures at CNU \$1,200	Fall 2021
The Excellence Prize of Capstone Design Contest at Engineering Innovation Center \$72	Sep. 2021
The Third Prize of Public Data Utilization Contest at Gwangju Metropolitan City \$360	Jul. 2020

Service & Activities

Reviewer at Springer Nature	Jun. 2024 – Present
IEEE Puzzlers Puzzler Quest Level - Silver	Mar. 2024 – Present
ICC DDINS Workshop Technical Program Committee (TPC) reviewer	Feb. 2025 – Mar. 2025
IEEE Graduate Student Member	Jan. 2024 – Dec. 2024

Technical Skills

Hardware Platforms: Raspberry Pi, LattePanda Mu, Jetson Orin Nano, Jetson Nano/Xavier, Arduino UNO
NPUs: Hailo-8/8L/10H (Hailo), Regulus/Aries (Mobilint), DX-M1 (DeepX)
Programming Tools: TensorFlow, PyTorch, OpenCV, Hugging Face, Roboflow, Docker, AWS, gnuplot, Node.js
Collaboration Tools: Overleaf, Github, Notion, Discord, Slack, Microsoft Teams
Languages: English, Korean, Japanese